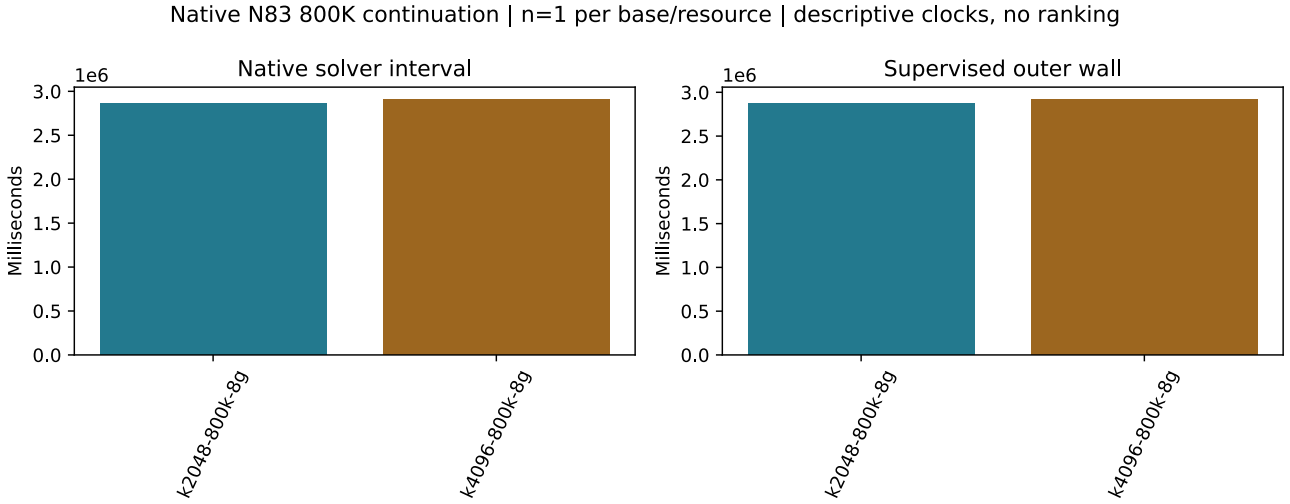
Native N83 sweep through 800000 conflicts

UTC round2026-10-20; retained single public query; K2048 and K4096.

Both corrected8GiB attempts completed800000conflicts with native bounded-zero replay: models0, relations0, rank0. Exit0, no OOM or wall censor, cleanup verified; exact cards/receipts/ stream digests and verbatim relays passed director assessment. These are bounded unknowns, not UNSAT, a best-base result or scalar recovery.

Run	Conflicts	Audit	Solver ms	Outer ms
k2048-800k-8g	800000	PASS bounded zero	2866110.318261	2871741.354358
k4096-800k-8g	800000	PASS bounded zero	2902661.855852	2913487.794923

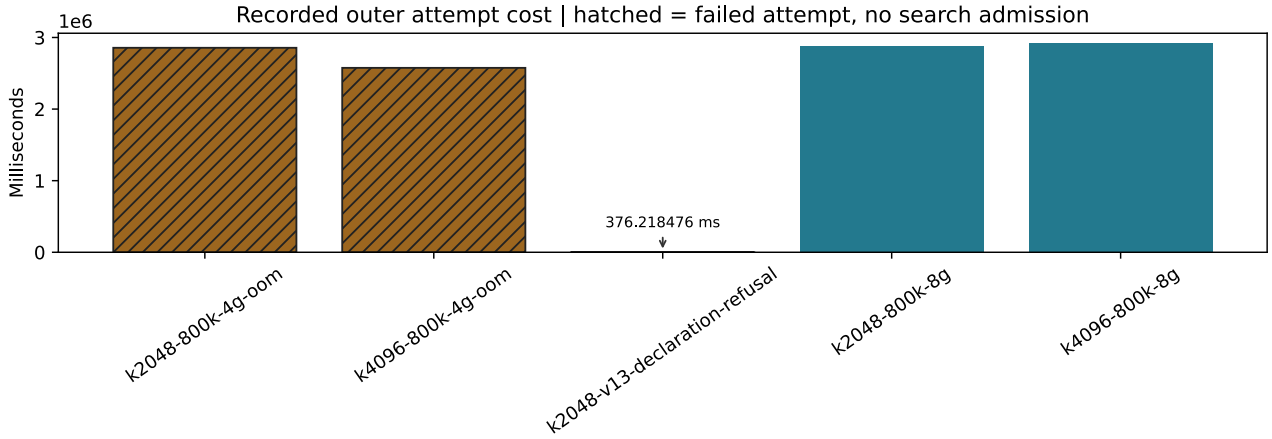


Native solver wall is solver_return minus solver_enter, calculated by the Rust summary helper. Outer wall is the recorded supervised interval. n=1 admitted per base/8GiB envelope; no uncertainty estimate or timing ranking is established. The guard does not grant automatic L2: fixed kernel threads and missing worker context-switch/fault counters remain. These descriptive clocks do not establish end-to-end IC improvement, a cold minimum runtime or a matched-rho ratio.

Preserved resource failures

Original4GiB v12 attempts were OOM-killed (exit137, cleanup verified): K2048 last progress789000conflicts, failed-attempt outer2856001.340487ms; K4096 last progress702000conflicts, outer2575759.292233ms. Both SAT traces are valid nonterminal prefixes. Terminal models, relations and rank remain missing. Full search admission stays refused; these costs are recorded failure costs, not admitted solve intervals.

K2048-v13 set Docker8GiB but still declared4096MiB to the worker. The worker refused before query/SAT start: exit1, noOOM, cleanup verified, outer376.218476ms, command10.146594709s. K4096-v13 was never compiled/executed. Guard-v14 derives 8192MiB from the frozen8GiBbytes and the corrected actual worker configs match. All three failures and the two completed attempts remain separate records.



Frozen workload and execution

Curve a=0 icv1-f2m83-tm6151469093347-debefd74; field x^83+x^45+x^2+x+1; subgroup241785163923079621668568 9. Same public_x_hash seed2026100801, m5/ identitymask0 and retained K2048/K4096 public panels. The positive-trace a=1 holdout is distinct; EC1 identity equivalence remains unresolved.

Original PROTOCOL.md freezes4GiB attempts; MEMORY-V13.md separately admits8GiB; MEMORY-V14.md corrects the worker declaration before further search. Serial K2048->assessment->K4096;800000conflicts/4800worker seconds/5400command seconds. Five executed attempts each had4800seconds maximum, so this round’s executing or newly planned envelope was24000seconds, including the immediate refusal’s full allocation. K4096-v13′s unused planned leg was superseded, never charged as an observed run. Setup, tests, replay, rendering and storage are separate costs; no historical campaign balance or reused unused allowance is inferred.

One CPU quota, full SMT core14/30 reserved/runCPU14; explicit4GiB or8GiB memory with MemorySwap equal memory/no extra swap, networknone/read-only/capdropALL, no-new-privileges/64PIDs. Native exclusive lock, strict topology/sibling/cgroup checks, quiet samples and movable-user-thread eviction/restoration are retained. No task builds/tests/rendering occurred within measured arms. Host Xeon6975P-C, 32logical/16cores/SMT2; resumed home uses EFS and a fresh Docker daemon. Exact image was restored by digest and same-image /bin/true readiness passed. Migration precludes treating older rounds as matched-host timing comparisons.

Source and validation

The worker was not rebuilt; SHA256 `3a31bd2bedee5519d30b517f39ec27f72561ce898213657688c55b51bbcdf7b6`. Nine watched worker sources match660f0f4b10. Corrected frozen guard source 786a8af0f7f935f3e17e9be21b8930eaad113c61; guard-v14SHA256 `02adb545e54f51f35988efcce7f9a5b1be9a83e08db1a91dc4e66f2f3647011b`. Guard22unit+7execution-card tests pass, including frozen memory/swap forgeries and the actual worker argument contract. Earlier v12/ v13 ELFs, source freezes, intermediate/final test logs and every actual card/relay/assessment are retained. Native replay independently checks the public query in the original field/group, phases, source/input/ELF/stream custody, resource settings, exit/OOM and cleanup. Fresh serial release root-library validation: test result: ok. 4407 passed; 0 failed; 118 ignored; 0 measured; 0 filtered out; finished in 644.23s Exporter19tests and independent model-checker2tests pass in the separate fresh validation target after timing. The retained measured worker ELF is unchanged. Earlier4398-pass/118-ignored and post-main-merge socket-denial/38-module-rerun logs remain historical evidence; this new run is a distinct clean result.

Canonical cold/rho/identity/timeline/leaderboard graphs checked and unchanged: no new speedup, curve identity, scalar recovery or end-to-end result is promoted. This report’s graphs and benchmark copies add bounded-search coverage only.

Evidence custody and remaining work

Study: research/koblitz_n83_factor_base_sweep_20261008. Protocol/addenda, native SUMMARY.json, compact records, cards/assessments and PUBLICATION.md: verification/native-natural-800k-20261020/. Source-linked report and editable SVG/same-source150dpiPNG: reports/run-20261020T0600Z/; benchmark copies: benchmarks/20261020-n83-natural-800k/. Exact ELF/public inputs/source archives and all streams belong to the raw bundle.

S3 readiness currently fails: direct conditionalPutObject is denied because the migrated AWS role lacks s3:PutObject for setec-astronomy-artifacts. No successful upload or fresh S3 GET is claimed. PUBLICATION.md records final exact archive bytes/hash, actual full-bundle transport outcome and the scope of local extraction, payload checks, native replay and deterministic repack. Local verification does not substitute for fresh S3 custody. Prior verified publications stay immutable.

Next discriminating work: profile learnt-database memory before another higher cap, retaining this worker/query and separately freezing any eviction policy or resource change. No further timed search is scheduled by this report. Useful natural rank, cold/rho performance and the original4GiB execution issue remain separate obligations. Current storage access needs restoration and fresh readback.